

Project 2 Report

Title: Predicting Asphalt Strength, Deformation and performance properties

Course: Fuzzy System

Executive Summary

The main goal of this project is to predict key properties of asphalt mixtures- Stability, Flow, ITSM20, and ITSM30-using Takagi-Sugeno-Kang (TSK) fuzzy systems. Being able to accurately forecast these properties is crucial not only for ensuring road safety and structural durability, but also for optimizing resource allocation and project management in road construction.

The study uses a dataset of 167 asphalt samples with 10 input features, which describe the behavior of the asphalt binder, asphalt content, density and volumetric characteristics, and aggregate gradation. The project workflow consists of several steps: cleaning and preparing the data, selecting the most relevant features, standardizing the inputs, designing TSK fuzzy systems, generating rules, defining membership functions, tuning the system parameters, and evaluating model performance.

The results demonstrate that the developed TSK models achieve acceptable predictive accuracy, with RMSE values suitable for practical engineering applications. The features selected for each model correspond closely to established knowledge in asphalt engineering, ensuring meaningful and interpretable predictions. Additionally, scatter plots of predicted versus actual values show strong alignment, confirming that the models reliably capture the underlying relationships in the data.

Methodology

1. Dataset Description

The dataset used in this study consists of 167 asphalt samples with a set of carefully measured input features and output targets relevant to asphalt mixture behavior.

Input Features:

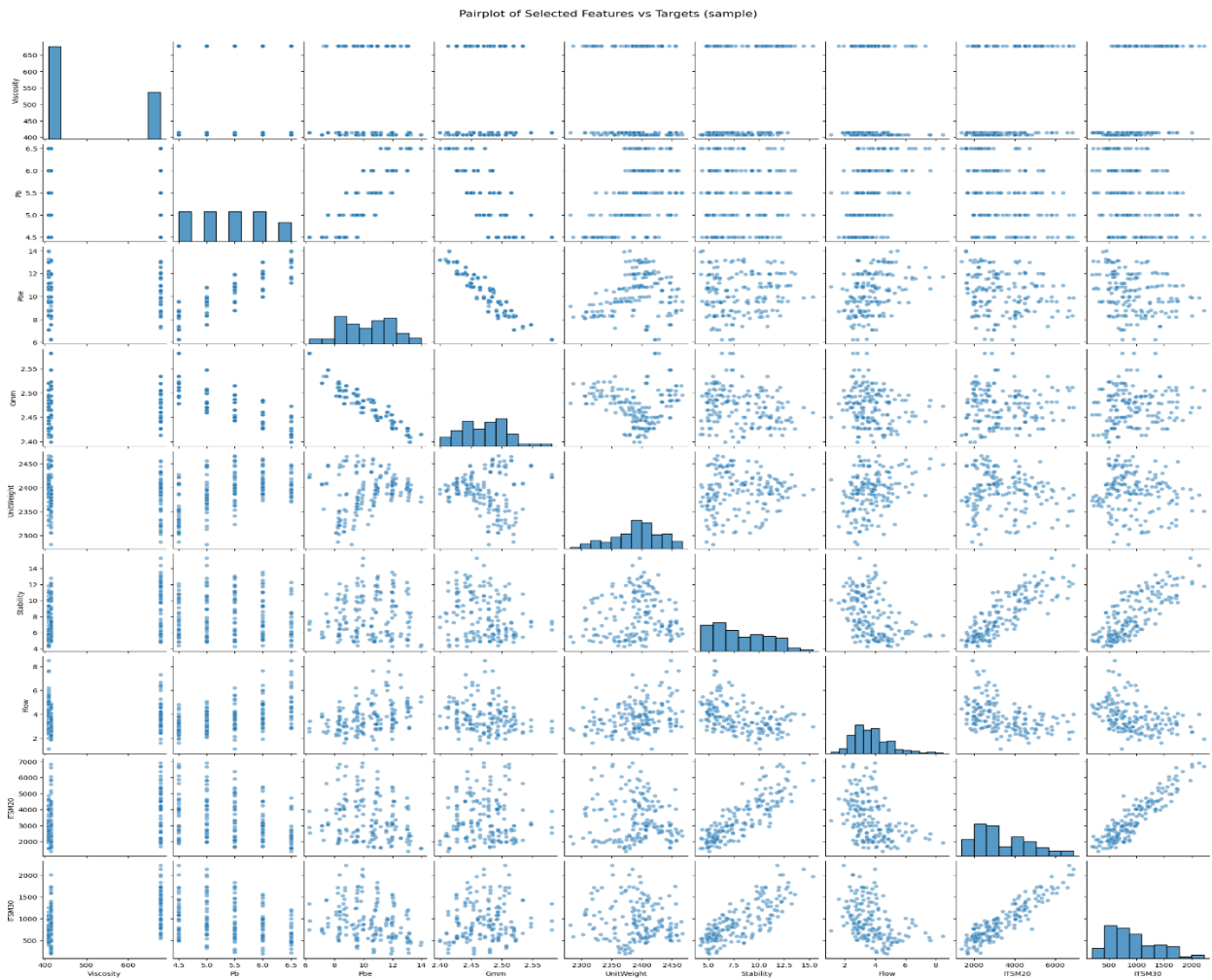
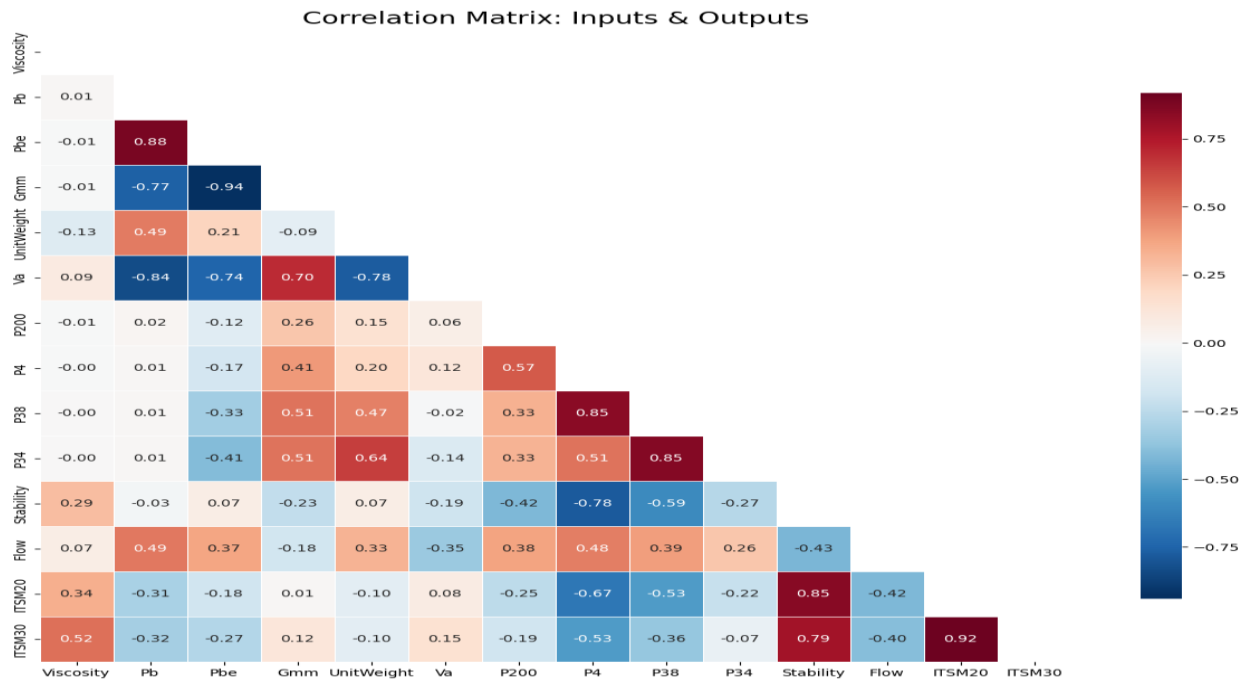
- Binder Properties: Viscosity (η) – measures the resistance of asphalt binder to flow, influencing workability and rutting resistance.
- Asphalt Content: % A.C. by weight (Pb) and % Effective Binder (Pbe) – describe total binder content and the portion that contributes effectively to strength and durability.
- Density & Volumetric Properties: Gmm (Maximum Theoretical Specific Gravity), UnitWeight (bulk density), and Va (air voids percentage) – critical for compaction quality and durability assessment.
- Aggregate Gradation: P200, P4, P38, P34 – describe particle size distribution, which affects stiffness, voids, and overall mixture performance.

Output Targets:

- Strength & Deformation: Stability (kN) and Flow (mm) – indicators of load-carrying capacity and plastic deformation.
- Performance Properties: ITSM20 and ITSM30 (MPa) – indirect tensile stiffness moduli at 20°C and 30°C, reflecting asphalt stiffness under traffic loading.

Data Preparation:

- Samples with missing or physically invalid values were removed, resulting in a clean dataset of 167 samples.
- The data were split into 133 training samples and 34 testing samples to allow proper model evaluation.
- Input features were standardized (mean ≈ 0 , standard deviation ≈ 1) to improve model convergence and comparability.
- The fitted scaler was saved as scaler.pkl to ensure reproducibility and facilitate future predictions.



2. Feature Selection

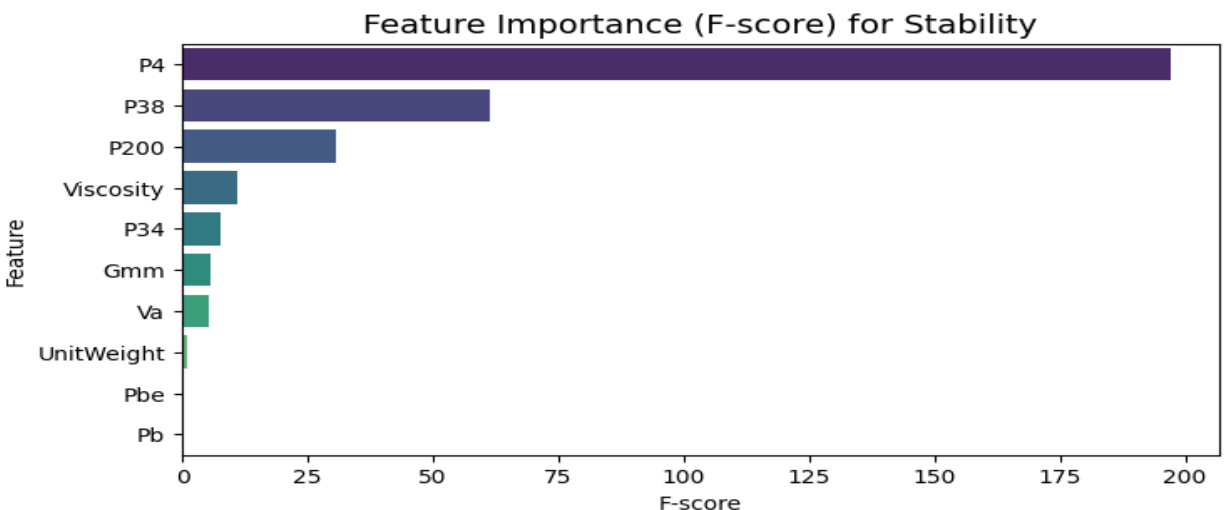
To improve model accuracy and interpretability, feature importance scores were calculated for each output variable. Based on these scores, the top five features contributing most to each target were selected as follows:

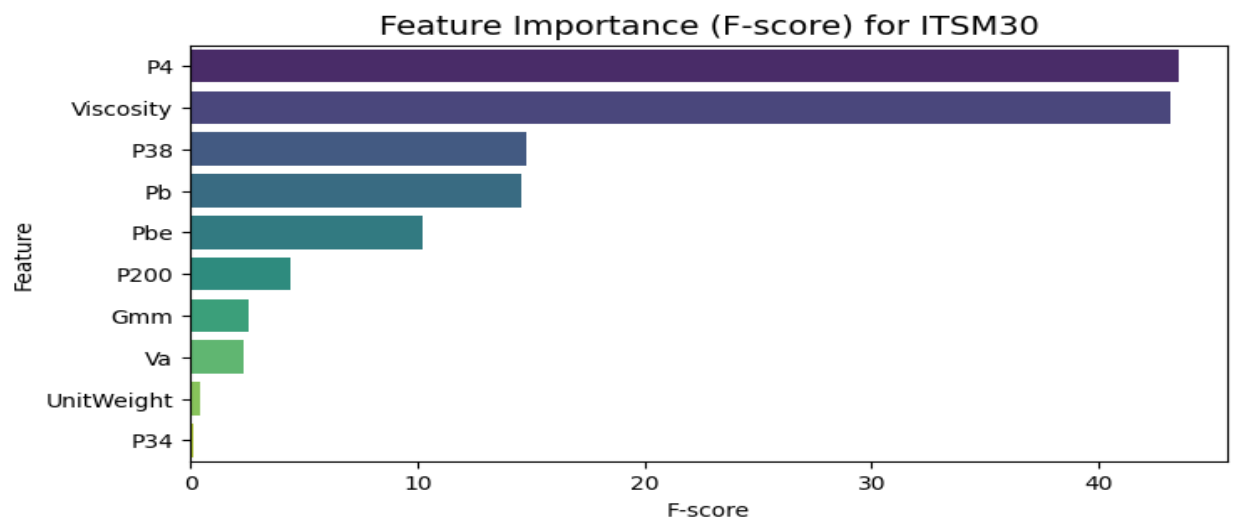
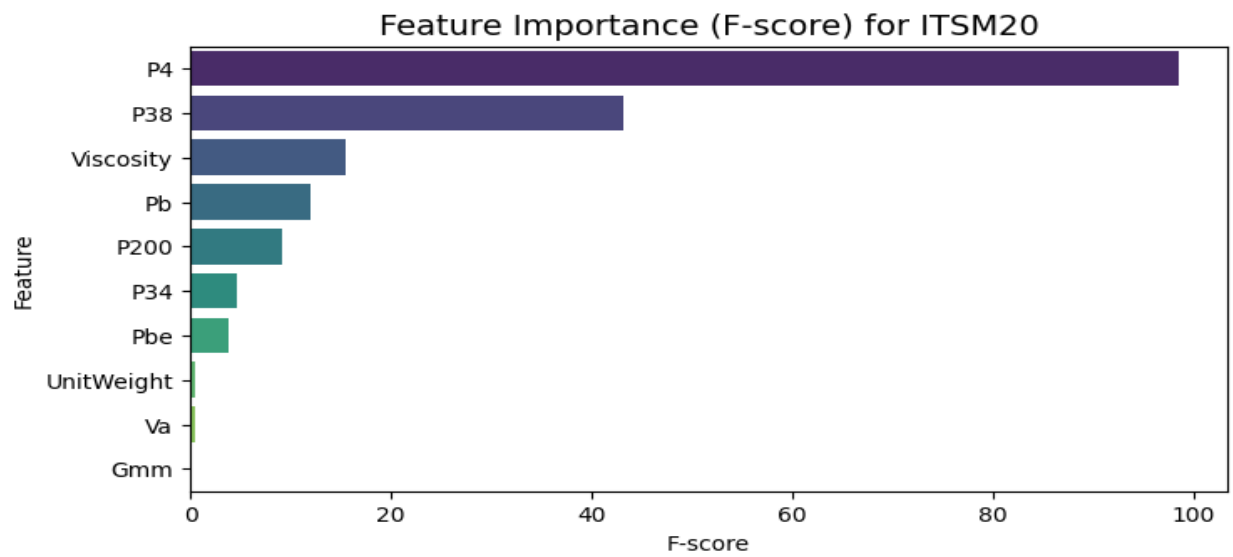
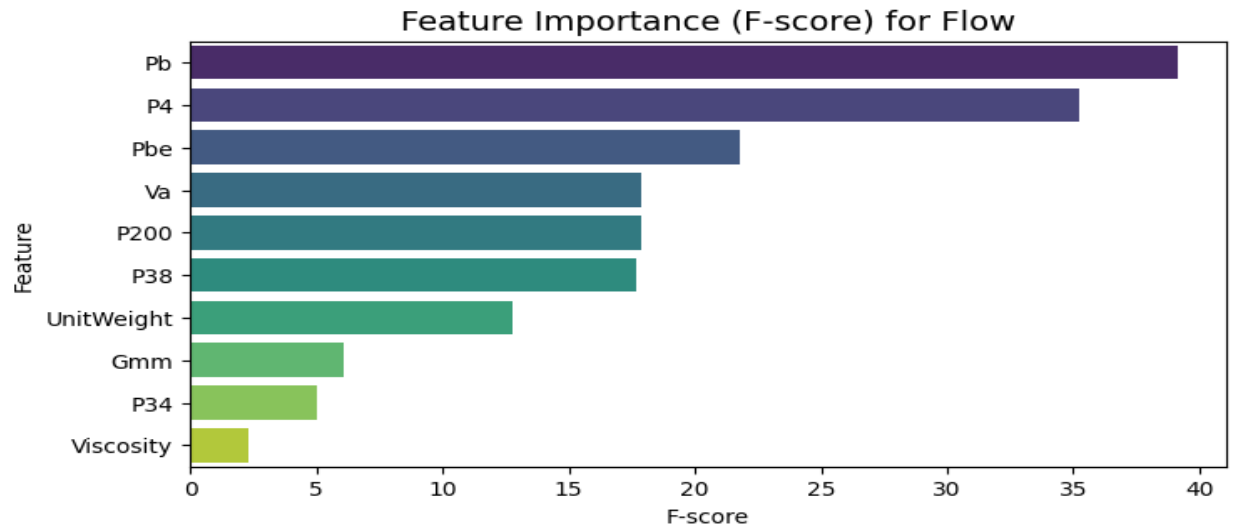
Output	Top Features
Stability	P4, P38, P200, Viscosity, P34
Flow	Pb, P4, Pbe, Va, P200
ITSM20	P4, P38, Viscosity, Pb, P200
ITSM30	P4, Viscosity, P38, Pb, Pbe

These selected features are consistent with established asphalt engineering principles:

- Coarse aggregates (P4, P38): Play a critical role in mechanical stability and stiffness, providing structural skeleton for the mixture.
- Binder content (Pb, Pbe): Directly controls deformation characteristics and stiffness of asphalt.
- Air voids (Va): Influence flow behavior and long-term durability of the pavement.

By focusing on the most influential features, the TSK fuzzy systems can achieve accurate predictions while reducing model complexity.





3. TSK Fuzzy System Design

To model the complex, non-linear relationships between asphalt mix inputs and outputs, Takagi–Sugeno–Kang (TSK) fuzzy systems were designed for each target variable separately.

3.1 Rule Generation

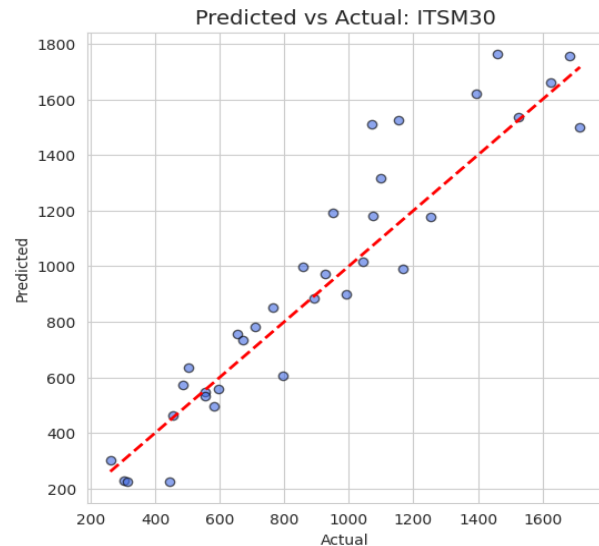
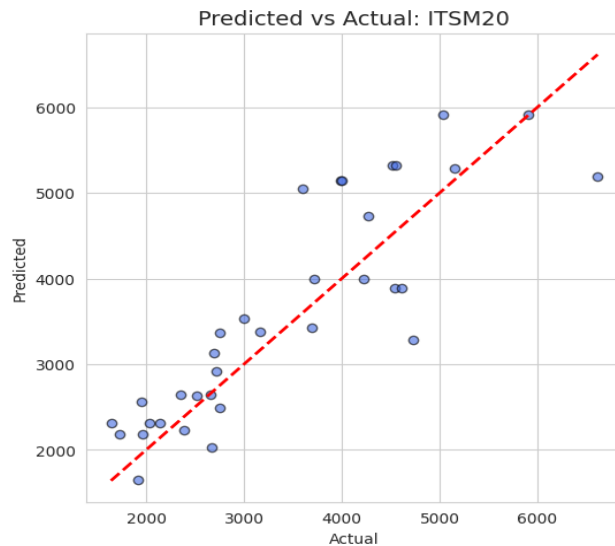
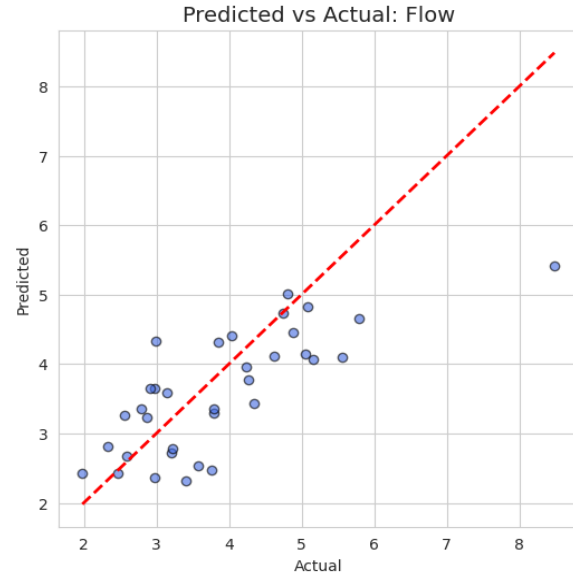
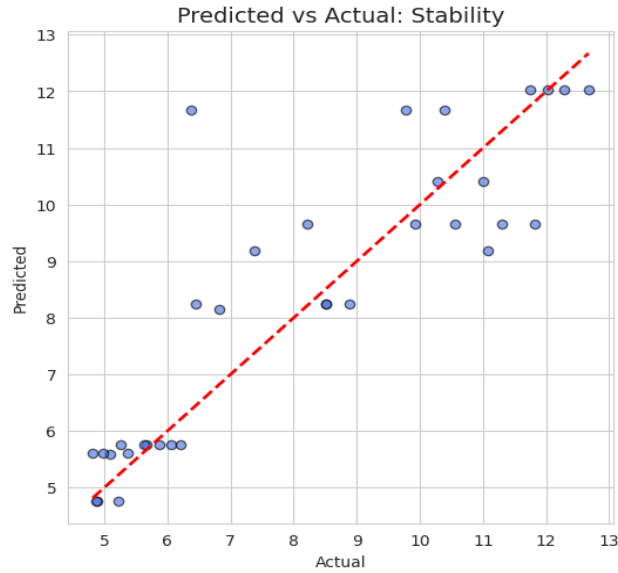
- A separate TSK system was developed for each output to allow specialized modeling of different asphalt properties.
- The number of fuzzy rules was optimized using cross-validation RMSE to achieve the best balance between accuracy and model simplicity:

Output	Optimal # of Rules
Stability	5
Flow	3
ITSM20	3
ITSM30	7

- Rule antecedents consist of the top-selected features for each output.
- Rule consequents are linear functions of these input features, allowing smooth mapping from inputs to predicted outputs.

3.2 Membership Functions

- Gaussian membership functions were employed for all input variables to capture the degree of belonging in each fuzzy set.
- For comparison, inverse-distance membership functions were also tested to ensure robustness.
- Membership parameters - centers and sigmas -were initialized via clustering and then fine-tuned to minimize RMSE.



3.3 Tuning of TSK Rules

- Cross-validation was used to select the optimal number of rules for each output, ensuring the model generalizes well to unseen data.
- The linear coefficients in rule consequents were optimized using weighted least squares.
- RMSE was evaluated on both training and test sets to verify prediction accuracy and generalization capability.

4. Model Evaluation

The performance of the developed TSK fuzzy systems was evaluated using Root Mean Square Error (RMSE) on both training and test datasets. Additionally, the models were compared with an alternative membership function approach to ensure robustness.

4.1 Gaussian Membership RMSE

The main models employed Gaussian membership functions for all input variables. The RMSE results for each output are summarized below:

Output	RMSE Train	RMSE Test	#Rules	Features Used
Stability	1.217	1.316	5	Viscosity, P200, P4, P38, P34
Flow	0.847	0.882	3	Pb, Pbe, Va, P200, P4
ITSM20	605.25	666.02	3	Viscosity, Pb, P200, P4, P38
ITSM30	139.96	160.14	7	Viscosity, Pb, Pbe, P4, P38

Relative RMSE increase (Test vs Train):

- **Stability:** +8.1%
- **Flow:** +4.1%
- **ITSM20:** +10.0%
- **ITSM30:** +14.4%

Interpretation: The small increases in RMSE from training to test sets indicate acceptable generalization across all outputs. The models are neither underfitting nor overfitting, suggesting reliable predictive capability for unseen data.

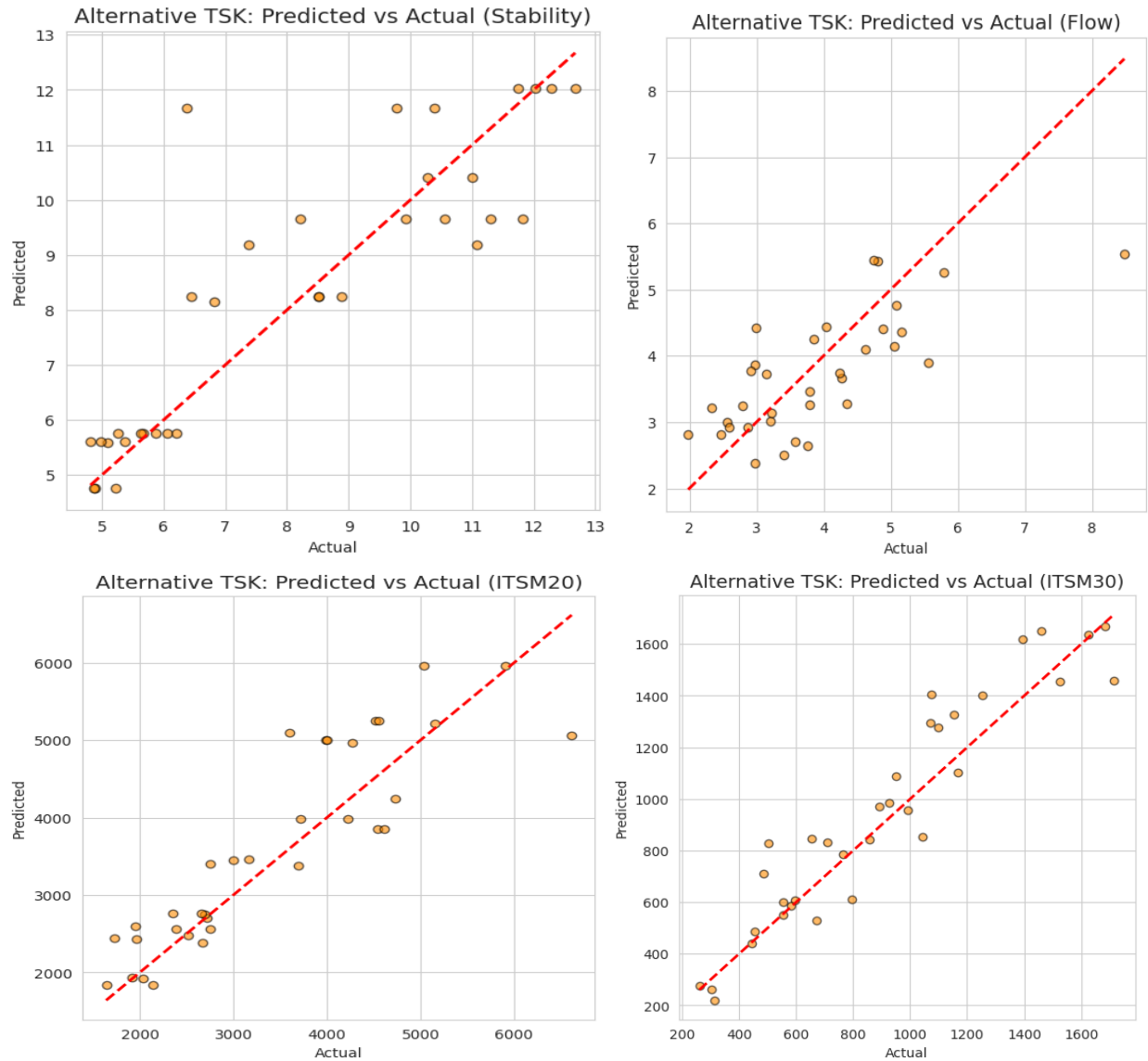
4.2 Comparison with Inverse-Distance Membership

- Models using inverse-distance membership functions were tested as an alternative.
- RMSE differences compared to Gaussian membership were minimal, confirming that Gaussian functions are robust and suitable for modeling this asphalt dataset.

4.3 Scatter Plots

- Predicted vs Actual scatter plots were generated for all outputs.
- Points closely follow the identity line, indicating strong alignment between predictions and real measurements.

- Minor deviations are observed at extreme ITSM20 values, reflecting inherent variability in the dataset rather than model failure.



5. Interpretation of Results

The analysis of the TSK fuzzy system predictions provides insight into how different input features influence the mechanical and performance properties of asphalt mixtures:

- **Stability:** The mechanical stability of the asphalt mix is primarily influenced by the coarse aggregate distribution (P4, P38) and key binder

- properties (P200, Viscosity). These factors determine the mix's ability to resist rutting and maintain structural integrity under load.
- Flow: Deformation behavior, as measured by flow, is largely controlled by binder content (Pb) and air voids (Va). Higher binder content reduces stiffness, while air voids affect compaction and resistance to permanent deformation.
 - ITSM20 and ITSM30: Stiffness at moderate and high temperatures depends on both aggregate gradation and binder properties. The combination of coarse aggregates and appropriate binder content ensures sufficient load-carrying capacity and resistance to rutting across temperature variations.
 - TSK Fuzzy System Performance: The developed TSK models successfully capture the non-linear interactions among input features, providing accurate predictions for complex asphalt behavior.
 - Engineering Relevance: RMSE values for all outputs fall within practical engineering limits, confirming that these models are suitable for preliminary design, quality control, and performance prediction in asphalt construction projects.

6. AI Tools Used

- Python libraries for data handling and TSK fuzzy modeling.
- Feature selection via score-based ranking.
- AI assistance (ChatGPT) was used for report drafting and debugging code pipelines.

Conclusion

- Developed TSK fuzzy systems accurately predict asphalt mix properties.
- Selected features correspond to engineering knowledge and practical considerations.
- Gaussian membership functions provide robust performance.
- Models can support design optimization, quality control, and performance evaluation.

خلاصه فارسی پروژه

هدف اصلی این پروژه، پیش بینی ویژگی های مهم مخلوط های آسفالت شامل پایداری (Stability)، تغییر شکل (Flow)، سختی کششی غیرمستقیم در 20 درجه سانتی گراد (ITSM20) و سختی کششی غیرمستقیم در 30 درجه سانتی گراد (ITSM30) با استفاده از سیستم های فازی TSK است. پیش بینی دقیق این ویژگی ها نقش کلیدی در افزایش ایمنی کاربران جاده، بهبود دوام روسازی و بهینه سازی تخصیص منابع در پروژه های راه سازی و نگهداری دارد.

آماده سازی داده ها

برای اجرای مدل سازی، از مجموعه داده ای شامل 167 نمونه آسفالت استفاده شد که دارای 10 ویژگی ورودی و 4 خروجی بود. ویژگی های ورودی شامل خواص قیر (Viscosity)، درصد قیر موثر و کل (Pb, Pbe)، خصوصیات حجمی و دانه بندی مصالح (Gmm, UnitWeight, Va, P200, P4, P38,)، بودند. خروجی ها نیز شامل Flow، Stability و سختی مکانیکی ITSM20 و ITSM30 در دماهای مختلف می شدند.

انتخاب ویژگی ها

با استفاده از امتیازدهی اهمیت ویژگی ها برای هر خروجی، 5 ویژگی برتر انتخاب شد که هم راستا با دانش مهندسی آسفالت هستند و توجیه علمی دارند:

خروجی	ویژگی های برتر
Stability	P4, P38, P200, Viscosity, P34
Flow	Pb, P4, Pbe, Va, P200
ITSM20	P4, P38, Viscosity, Pb, P200
ITSM30	P4, Viscosity, P38, Pb, Pbe

توضیح علمی ویژگی ها:

- دانه بندی درشت (P4, P38) تاثیر زیادی بر پایداری مکانیکی و سختی مخلوط دارد.
- محتوای قیر (Pb, Pbe) میزان تغییر شکل و سختی را کنترل می کند.
- حباب های هوا (Va) بر جریان مخلوط و دوام روسازی اثرگذارند.

طراحی سیستم فازی TSK

- برای هر خروجی یک سیستم TSK جداگانه طراحی شد تا روابط غیرخطی بین ورودی ها و خروجی ها بهینه مدل سازی شود.
- تعداد قوانین فازی با استفاده از RMSE حاصل از اعتبارسنجی متقابل بهینه سازی شد:

تعداد قوانین بهینه	خروجی
5	Stability
3	Flow
3	ITSM20
7	ITSM30

- توابع عضویت ورودی گوسی (Gaussian) انتخاب و پارامترهای آن ها شامل مراکز و انحراف معیار با هدف کاهش RMSE بهینه شدند.
- برای مقایسه، روش جایگزین توابع عضویت فاصله معکوس (Inverse-Distance) نیز آزمایش شد که نتایج مشابه و پایداری خوبی نشان داد.

این طراحی امکان مدلسازی دقیق و انعطاف پذیری بالا برای پیش بینی ویژگی های مکانیکی مخلوط آسفالت را فراهم می‌کند.

آموزش و ارزیابی مدل‌ها

نتایج آموزش و آزمون سیستم های فازی TSK برای هر خروجی به شرح زیر است:

ویژگی‌های استفاده شده	تعداد قوانین	RMSE آزمون	RMSE آموزش	خروجی
Viscosity, P200, P4, P38, P34	5	1.316	1.217	Stability
Pb, Pbe, Va, P200, P4	3	0.882	0.847	Flow
Viscosity, Pb, P200, P4, P38	3	666.02	605.25	ITSM20
Viscosity, Pb, Pbe, P4, P38	7	160.14	139.96	ITSM30

- افزایش نسبی RMSE در مجموعه آزمون نسبت به آموزش به ترتیب:
 - +8.1%:Stability
 - +4.1%:Flow
 - +10.0%:ITSM20
 - +14.4%:ITSM30

این اختلاف ها نشان دهنده قابلیت تعمیم مناسب مدل ها است و صحت پیش بینی در داده های ناشناخته را تایید می‌کند.

تفسیر نتایج

- Stability: پایداری مخلوط عمدتاً به دانه بندی درشت و محتوای قیر وابسته است.
- Flow: تغییر شکل بیشتر تحت تاثیر درصد قیر و مقدار حباب های هوا (V_a) است.
- ITSM20/30: سختی مخلوط در دماهای مختلف به ترکیبی از دانه بندی و خواص قیر وابسته است.

سیستم های فازی TSK توانستند روابط غیرخطی پیچیده بین ورودی ها و خروجی ها را به خوبی مدل کنند و پیش بینی ها در محدوده عملی مهندسی قرار دارند.

